



Pregnancy and Dental Health

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OMFS 513: Integrated Biomedical and Clinical Sciences 2

Mission: To advance the science and art of healthcare through education, service, research, personal wellness, and social accountability.

Learning Objectives & Competency Statements

Learning objectives

1. Describe the systemic changes noted in the body and oral cavity during pregnancy.
2. Review health recommendations prior to, during, and after pregnancy

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

Preparing for Pregnancy

- Medical and psychiatric/behavioral conditions
- History or current substance abuse
- Genetic screening
- Oral health
- Diet
- Interpersonal relationships
- Living conditions

Preparing for Pregnancy

- Assessment of medical and psychiatric health
 - Chronic medical conditions
 - Hypertension, diabetes, thyroid disease, and mental conditions
 - Current medications
 - Immunizations
 - Infectious disease exposure
 - STIs
 - Communicable and environmental pathogens
- Genetic screening

PREGNANCY & SUBSTANCE ABUSE

Pregnant women need specialized care to manage the unique risks they face when recovering from drug addiction. While drug use during pregnancy is hazardous to the woman and her fetus, improperly managed detox carries similar risks.

1. THE PREVALENCE OF OPIOID USE DISORDER

during pregnancy more than doubled between 1998 and 2011, to about **4 per 1,000** deliveries



2. WHAT IS THE RISK OF USING WHILE PREGNANT?

- Preeclampsia
 - Miscarriage
 - Preterm Delivery
 - Stillbirth
 - NAS
 - Low Birthweight
 - Birth Deformities
-

3. WHAT ARE THE TREATMENTS FOR OPIOID ABUSE?

- Abruptly quitting opioids during pregnancy can have several harmful consequences including: **Miscarriage, Premature Labor, Fetal Distress.**
- Many addiction treatment centers are not equipped with the resources or knowledge to adequately address the health of both the pregnant mother and her unborn child during rehab. Finding a facility that does know how to treat pregnant women can save the life of the mother and her unborn child.

Diet and Oral Health

- A healthy diet helps to create a healthy mouth
- Poor dietary habits can lead to tooth decay and other oral diseases
- Healthy diet:
 - Chesses, nuts, fish, fresh fruits and vegetables
 - Water
- What not to eat:
 - Foods with high amounts of sugars
 - Highly acidic snacks and drinks

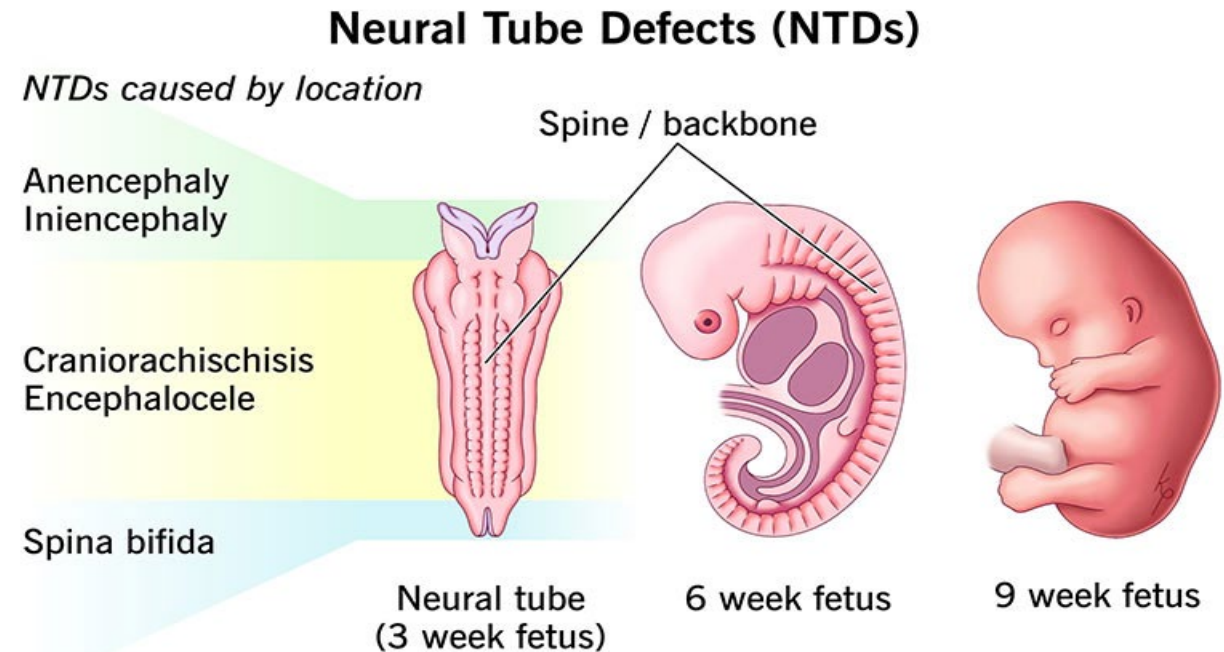


Preparing for Pregnancy - Supplements

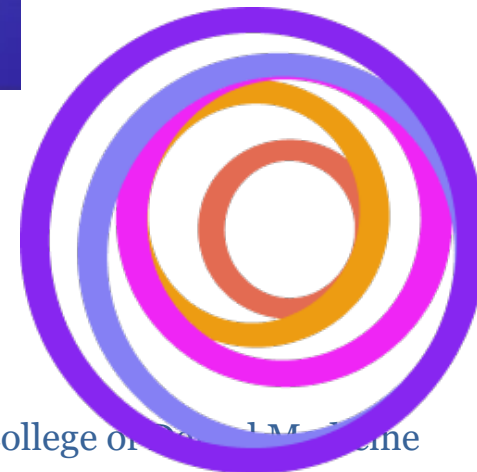
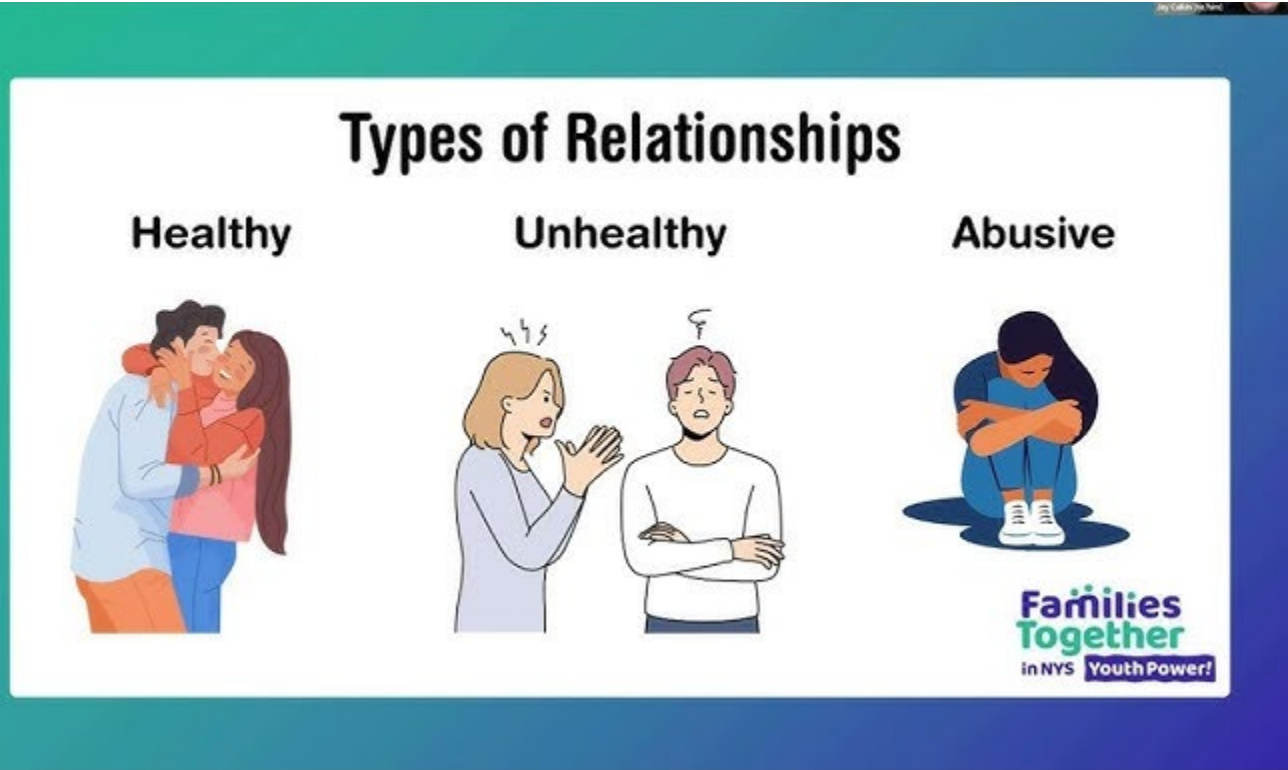


FOLIC ACID+ WITH VITAMIN B12 & IODINE

- Pregnancy Care
- Maintain Red Blood Cell Production
- Mental Health
- Normal Thyroid Function
- Immune Function
- General Wellbeing

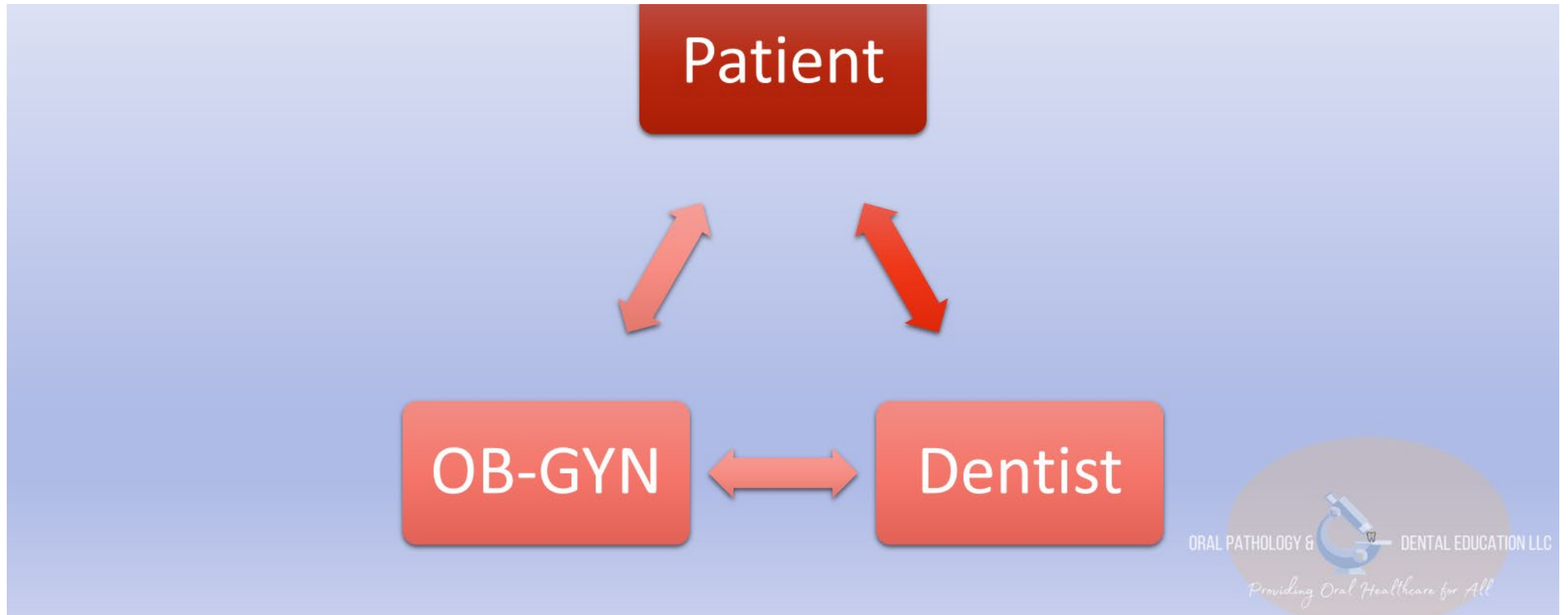



Relationships and Living Conditions

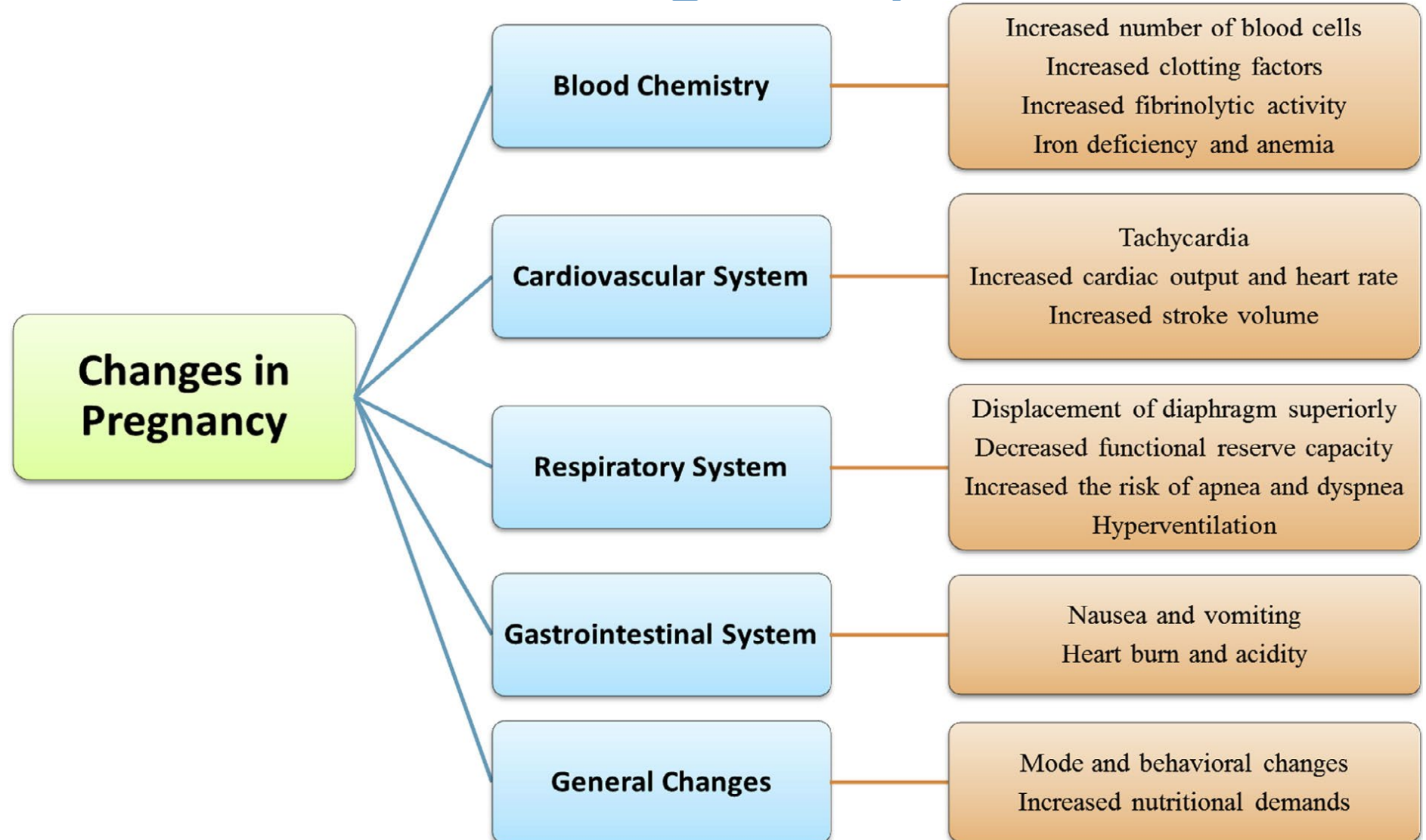


Homeless Prenatal Program

Pregnancy



Systemic Health Changes During Pregnancy



Changes in the mouth during pregnancy

- Pregnancy gingivitis
- Pyogenic granuloma “Pregnancy Tumor”
- Periodontal disease/odontogenic infections
- Dental erosion
- Dental caries

Changes to the Gingiva

- Pregnancy gingivitis
 - Increase in size of the gum tissue
 - Increase in blood vessels
 - Change in color - erythema
 - Softer texture/consistency
 - Change in bacterial composition



Changes to the Gingiva



Changes to the Gingiva

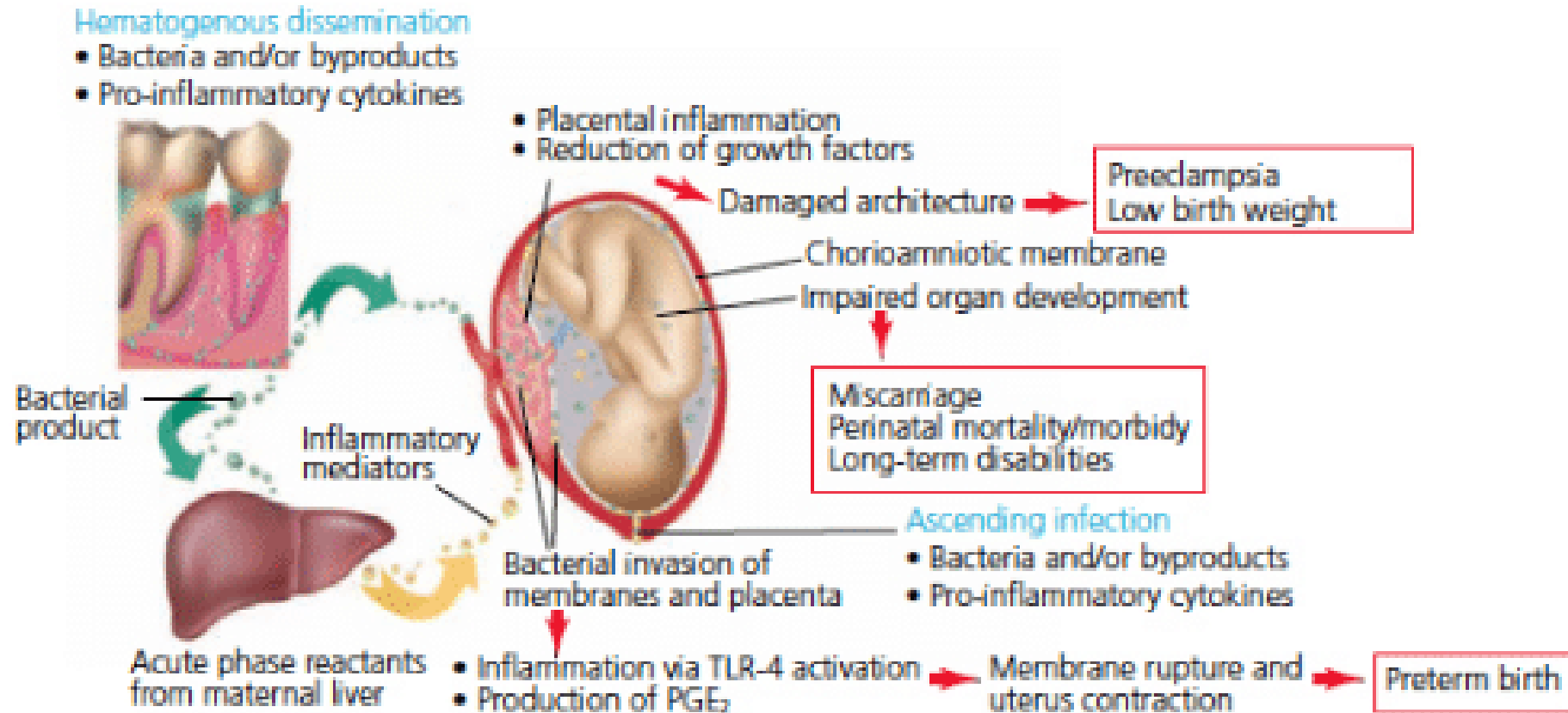
- “Pregnancy Tumor”
- Causes
 - Hormonal changes
 - Local irritation or trauma
- Smooth or lobulated vascular mass
- Commonly found on the gingiva
- Other sites: Tongue and buccal mucosa



Periodontal Disease

- Hormonal changes
 - Increased inflammation
 - Decrease immune regulation
 - Bone and tooth loss
- Link to adverse pregnancy outcomes
 - Pre-eclampsia
 - Gestational diabetes
 - Theoretical – preterm, low birth weight infants

Theory: Connection between periodontal disease and fetal adverse outcomes



Odontogenic Infections and Pregnancy

- Odontogenic infections in pregnancy
 - Infections can develop quickly and spread
 - Orofacial swellings
 - Compromise airway
 - Multiple procedures to clear infection
 - Untreated infections
 - Premature birth
 - May result in sepsis and miscarriage



Changes to Tooth Structure

- Dental Erosion

- Acidic oral environment

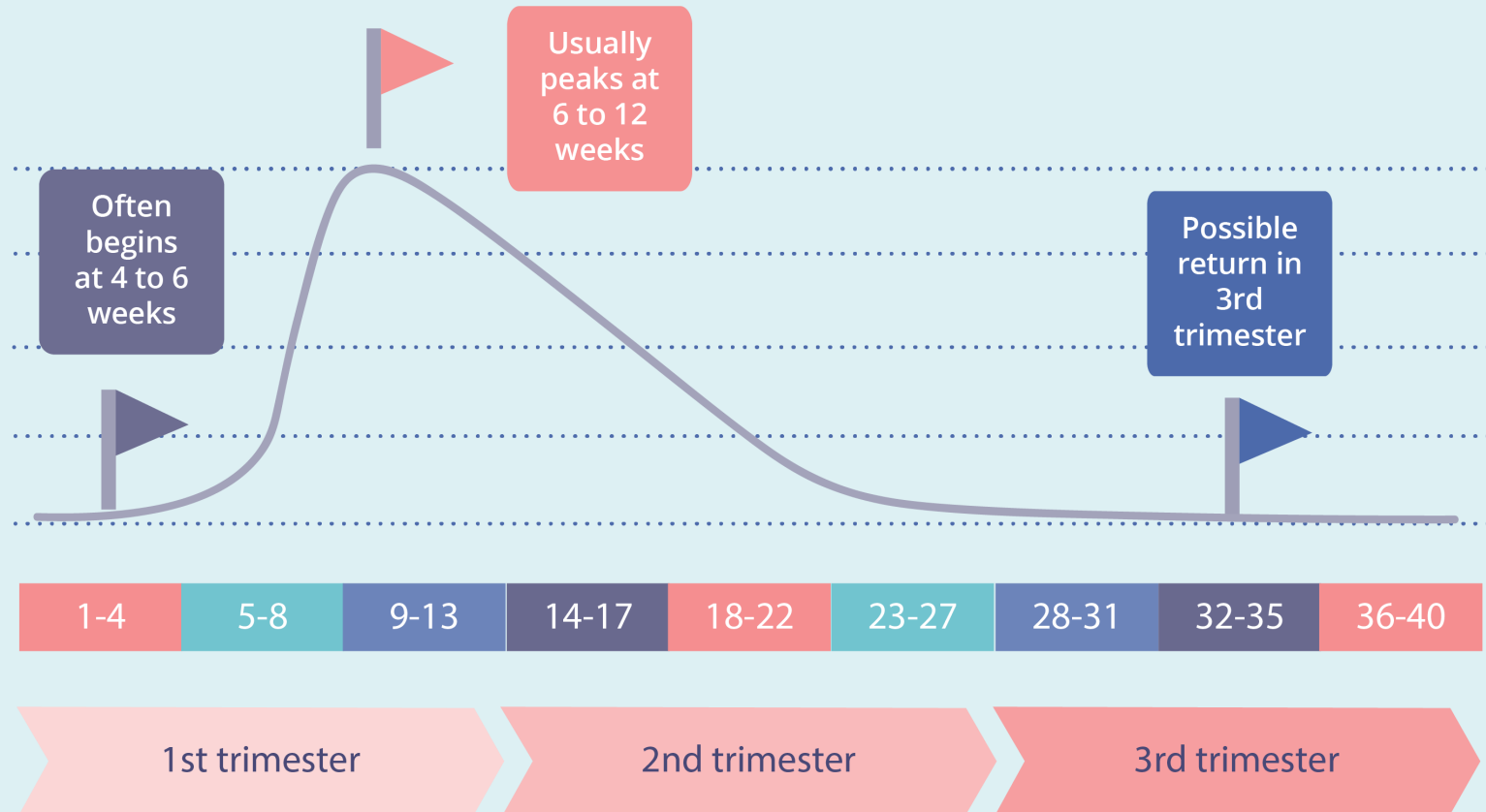
- Morning sickness/nausea
 - Acid reflux

Damage to dental enamel + sugary diets associated with food cravings



Tooth sensitivity and decay

Morning sickness throughout pregnancy



Combating Dental Erosion

- Antiemetics
- Rinse with baking soda and water
- Avoid brushing immediately after vomiting
- Use of soft bristle toothbrush and fluoride

Dental visits during pregnancy

- **Going to the dentist is highly encouraged during pregnancy**
- Emergency and preventive care can occur during each trimester
- Be mindful of local anesthesia and prescribed medications
- Best time for dental care is during the second trimester
- Radiographs can be taken with precautions
 - Take only necessary radiographs (ALARA principle)
 - Use a lead apron with a thyroid collar

Table 1 FDA risk categories of drug used during pregnancy and their potential risk factors.

Category	Risk factors	Antibiotics	Analgesics	Sedative hypnotics	Local anesthetics
A	Satisfactory well controlled studies on humans showing no hazard to the fetus				
B	Studies on animals demonstrating no fetal risk whereas no well controlled and adequate studies done on pregnant women	Amoxicillin Cephalexin Chlorhexidine Clindamycin Erythromycin Metronidazole Penicillin	Acetaminophen Ibuprofen		Lidocaine Prilocaine Prednisolone
C	Studies on animals establishing fetal hazards no controlled studies on human beings	Ciprofloxacin	Codeine with acetaminophen Hydrocodone + acetaminophen Propoxyphene		Mepivacaine
D	Evidence of risk to the fetus, can be used in exceptional cases or circumstances	Doxycycline Tetracycline (D)	Ibuprofen	Barbiturates Benzodiazepines	
X	The hazards of using the drug in pregnant women far more than the benefits	Nitrous oxide (avoided in the first trimester as it may result in neonatal depression and spontaneous abortion)			

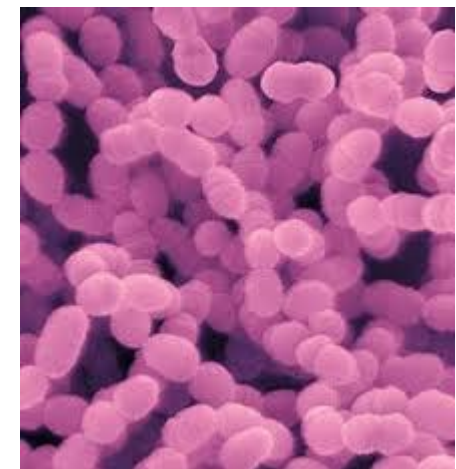
Postpartum

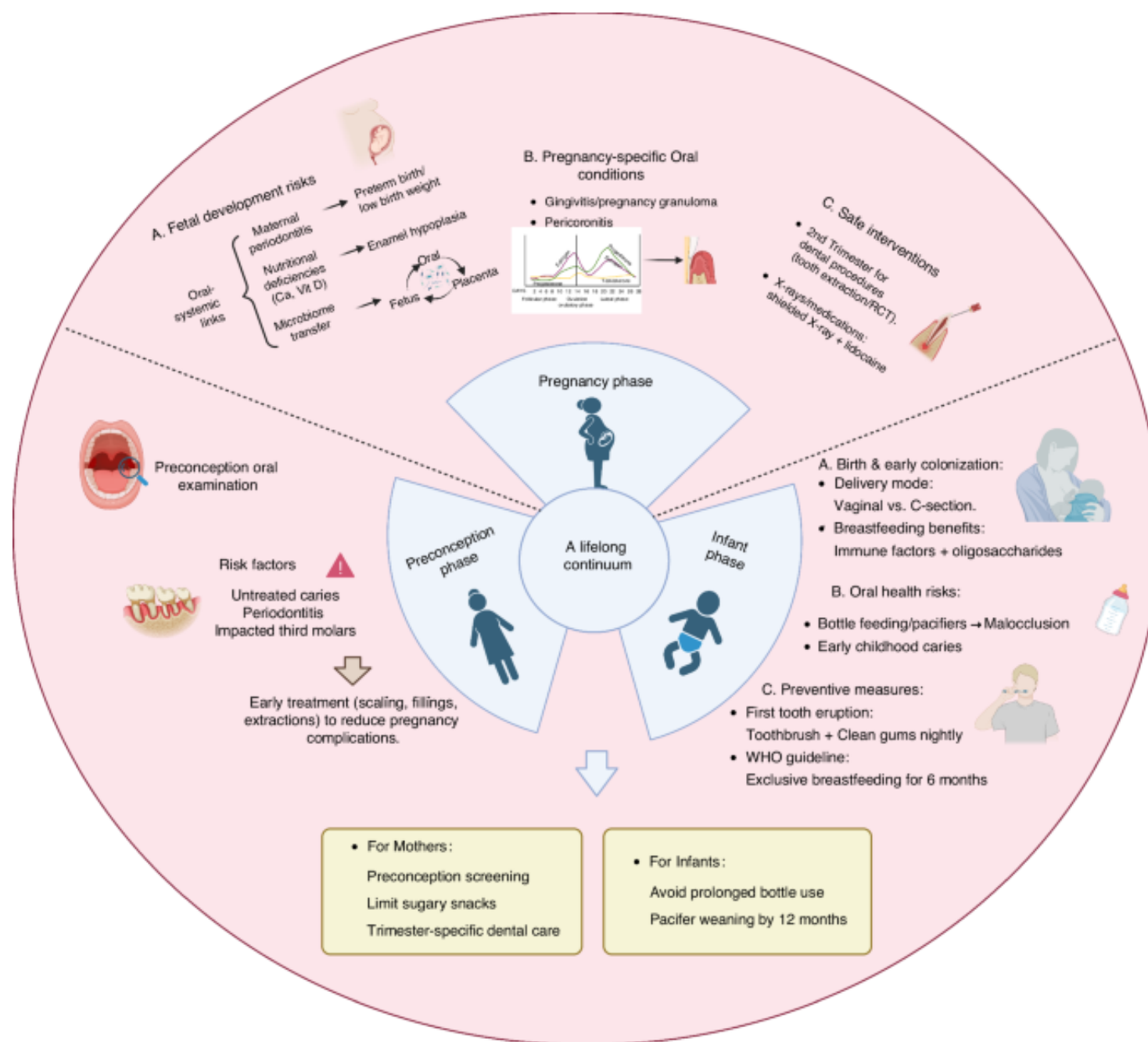
- Oral health concerns
- Mental health
 - Post-partum depression
- Chronic cardiovascular disease
- Medications and breastfeeding

Oral Hygiene for the Family

- Parents are teachers/role models
 - Teach them how to brush and floss
 - Dental visits twice a year starting a 1 year of age
 - Prepare child for first dental visit
 - Be mindful of diet







Resources

- Prepregnancy counseling. ACOG Committee Opinion No. 762. American College of Obstetricians and Gynecologists. *Obstet Gynecol* 2019;133:e78–89.
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